

Project 1: Overview, Presentation, and Grading

CIS 6270: Discrete Generative Models

Fall 2026

Project assigned: Thursday, September 10.

Code, paper, and presentation slides due: Tuesday, September 29, at 8:30 am. Submission instructions will be posted on Canvas.

Project defenses: Wednesday, September 30. (*Note: no slide changes are permitted after slide submission.*)

Paper format: NeurIPS workshop format, 5-page main-text limit. References and Appendix do not count toward the limit.

Total project score: 15 points: 7.5 points for the paper and code, and 7.5 points for the defense.

Project Overview

Project 1 is an empirical project on continuous generative modeling. Each team will choose **two continuous modalities**. Examples include images, audio, protein structures, 3D molecular coordinates, point clouds, continuous latent representations, and probability-simplex representations. Probability-simplex representations of discrete modalities count as a continuous modality.

For **Modality 1**, train one continuous flow-matching model and one diffusion model on the same generation task. Compare them under the same evaluation protocol, then show that at least one guidance mechanism works. Choose the stronger model family for your task and add a methodological innovation. Use ablations to show whether the innovation is actually responsible for the improvement. Finally, compare the proposed method against at least three relevant methods published within the last five years.

For **Modality 2**, transfer the *same core innovation* to a different continuous modality and compare it against at least three recent methods. You do not need to repeat the flow-matching-versus-diffusion comparison, the standard guided-versus-unguided comparison, or the Modality 1 innovation ablations.

The point of the project is to learn how to make and support a scientific claim. Your experiments should answer a sequence of questions: Which base model should you build on? Does guidance work? Does the proposed change help? Is the final method competitive with recent work? Does the same idea transfer to another modality?

You will receive a separate **Project 1 paper template**. Please follow it closely. It is both a paper template and a guide to the experiments expected in this project. It tells you what belongs in each section, what evidence should be shown, and gives examples of modalities, datasets, baselines, and possible innovations. The **main paper is limited to 5 pages before the references**. References and the Appendix do not count toward the limit.

Paper and Code (7.5)

Abstract

.225 points

Summarize the complete study in one compact paragraph. State the problem, two modalities, selected model family, guidance experiment, innovation, strongest result, Modality 2 transfer, and final conclusion.

1. Introduction

.525 points

Include the following:

- Motivate the generation problem and explain why flow matching and diffusion are relevant. (.1 pt)
- Identify the limitation or opportunity that motivates your innovation, and cite the closest prior work. (.225 pts)
- State the two-modality study design and the main hypothesis you will test. (.1 pt)
- End with concise contribution bullets that match what you actually show in the paper. (.1 pt)

2. Related Work

0.425 points

- Cover the flow-matching and diffusion work most relevant to your setup. (.1 pt)
- Discuss the papers most closely related to your proposed innovation. (.125 pt)
- Include the modality-specific literature and benchmark methods you compare against. (.1 pt)
- Make clear what is already known and what is different in your approach. (.1 pt)

3. Methods

2.575 points

3.1 Modalities, Data, and Preprocessing

(.3 pt)

State both modalities and datasets. Report the representation, train/validation/test split, preprocessing, conditioning information, and anything else needed to understand the two generation tasks. Explain why the two choices are genuinely different continuous modalities.

3.2 Continuous Flow Matching Baseline

(0.475 pt)

Describe the Modality 1 flow-matching model you actually trained: probability path, training target/objective, architecture, important training settings, and sampling procedure. The equations should match the implementation.

3.3 Diffusion Baseline

(.475 pt)

Describe the Modality 1 diffusion model you actually trained: forward process, prediction target/loss, architecture, important training settings, and reverse-time sampling procedure. The equations should match the implementation.

3.4 Guided Generation

(.3 pt)

State the guidance objective and explain exactly where the guidance signal enters generation. Report the guidance strength/schedule and any auxiliary model used. Make clear which components are trained, pretrained, or frozen.

3.5 Proposed Innovation

(.625 pt)

This is the main Methods subsection. State the problem you are trying to fix, explain the idea, and define the new mechanism clearly enough to reproduce it. Show how it differs from the selected base model and from the closest prior work. Also state which ablations will test whether the new component is responsible for the gain. Ordinary hyperparameter tuning, more epochs, more compute, a different random seed, or simply replacing a standard backbone with a larger standard backbone does not count by itself as the required innovation.

3.6 Transfer to Modality 2

(.2 pt)

Explain what stays conceptually the same when the innovation is transferred to Modality 2 and what must change because the data representation or model architecture is different.

3.7 Model Overview

(.2 pt)

Include one clear, publishable overview figure showing the Modality 1 development pipeline and how the same

innovation is transferred to Modality 2. Label trainable, pretrained, and frozen components when relevant.

4. Results

2.9 points

Organize the Results around the questions below. For each comparison, state the setup, show the result, and explain what it means.

4.1 Evaluation Protocol

(.4 pt)

Choose appropriate metrics for both modalities and explain what they measure. State the evaluation protocol, sampling/compute budget where relevant, number of seeds, and how uncertainty is reported. Keep comparison conditions matched whenever possible; if they are not matched, say so.

4.2 Modality 1: Flow Matching vs. Diffusion

(.45 pt)

Compare the two base models on the same task and evaluation protocol. Report generation quality/diversity and sampling efficiency where meaningful. Use the result to justify which model family you carry forward.

4.3 Modality 1: Guidance

(.325 pt)

Compare guided and unguided generation under otherwise matched settings. Show that the target property or condition improves, and also report at least one general quality/diversity/validity metric so that guidance is not judged by the target score alone. If the target property/condition does *not* improve from unconditional, investigate the reason and discuss your findings.

4.4 Modality 1: Innovation Ablations

(.675 pt)

This is the main evidence for the innovation. Compare the final method against the same base model without the innovation. If the innovation has multiple pieces, remove or modify the important pieces one at a time. The ablations should make it possible to tell which part is helping and whether there are trade-offs.

4.5 Modality 1: Comparison with Recent Methods

(.45 pt)

Compare the final method against at least three relevant methods published within the last five years. Use the same task and evaluation protocol whenever possible. Explain why each baseline is appropriate. Report cases where your method is better, similar, or worse rather than discussing only favorable metrics.

4.6 Modality 2: Transfer

(.4 pt)

Apply the same core innovation to Modality 2 and compare against at least three appropriate recent methods. You do not need to rerun the Modality 1 FM-vs.-diffusion, standard guidance, or innovation-ablation sequence. Report whether the innovation still helps and what had to be adapted for the new modality.

4.7 Cross-Modality Analysis and Failure Modes

(.2 pt)

Compare what happened across the two modalities. Report at least two meaningful limitations or failure modes. State what the experiments support about the innovation, where it does not help, and how general you think the result is.

5. Discussion

.325 points

- Summarize the main findings without overstating what the experiments establish. **(.125 pt)**
- Give a plausible explanation for why the innovation helped or failed, tied to the model, objective, geometry, guidance, or data as appropriate. **(.1 pt)**

- Discuss limitations and useful next steps. Keep the final claim within what the two-modality evidence supports. (.1 pt)

Appendix

There is no separate Appendix score. Use the Appendix for full hyperparameters, extra ablations, additional samples, complete benchmark tables, pseudocode, and implementation details. Evidence needed for a central claim should still appear in the main five pages.

Code Quality and Documentation (.525 points)

Submit the code used for the experiments in the paper. The code does not need to be a production software package, but it should be organized and understandable.

- **README and setup:** explain the repository structure, environment/setup, dataset preparation, and how to run training, sampling, evaluation, and figure generation. (.2 pts)
- **Organization and comments:** keep configs and experiment settings organized; comment non-obvious code, especially the implementation of your innovation. (.225 pts)
- **Connection to the paper:** identify which scripts/configurations produced the main experiments and cite external code you adapted. (.1 pt)

Defense (7.5 points)

Each team will have a **30-minute defense**: a **15-minute presentation** followed by **15 minutes of Q&A** with the teaching team.

Presentation

4.5 points

The presentation has a hard 15-minute limit. Students MUST follow the presentation sequence below.

General requirements

1. **Labeled slides:** Your slides must progress in order from 1a-3e. You may use more than one slide per item. The appropriate label (1a-3e) must be shown in the top-right corner of each slide.
2. **On-slide citations:** All sources must be cited.
 - (a) Use numbered citations in a consistent style of your choosing (*e.g.* “Diffusion models are great [1]” or “Flow matching models are awesome²”.)
 - (b) On the bottom-right corner of the slide, in a very small font, please briefly list sources consistent with your numbering format (*e.g.* “[1] Chatterjee et al. 2026”, or “²Chen et al. 2025”).
 - (c) Citations are required for all intellectual contributions *and* images derived from other sources. If there is any uncited figure on your slides, it must be original.
 - (d) You may choose to start with fresh numbering (1, 2, ...) on each slide, or not.
3. **Bibliography:** You must have a Bibliography slide at the end of the presentation. Use a consistent citation style of your choice. References do not have to appear in the order they were mentioned on slides.
4. **Evenly distributed talking time:** Students must present roughly equal amounts of content. Teams will be penalized if certain members dominate the presentation while others barely speak.

5. **No notes:** Students are *NOT* allowed to have flashcards or notes to aid in their presentations.

Part 1: Set Up the Study

1.3 points

1a. Frame the study

(.1 pts)

- **Problem:** State the generation problem and a central question or hypothesis. (.05 pt)
- **Modalities:** Introduce both continuous modalities and describe their state spaces. (.05 pt)

1b. Define the data and representations

(.275 pts)

- **Datasets:** Report both dataset sources, train/validation/test sizes, split strategy, and preprocessing; explain how leakage is prevented. (.1 pt)
- **Representations:** Report and justify the representations used. (.075 pt)
- **Conditioning:** Report and justify any conditioning used. (.05 pt)
- **Modality-specific handling:** Explain normalization, invariance handling, or simplex mapping and decoding as applicable. (.05 pt)

1c. Explain the flow-matching baseline

(.275 pts)

- **Mathematical formulation:** Define the implemented probability path, target velocity, loss, and new symbols; cite the relevant flow-matching literature. (.125 pt)
- **Architecture and training:** Report the architecture, optimizer, and important training settings. (.05 pt)
- **Model selection:** Explain how the final model was selected, using validation-based selection or early stopping where appropriate. (.05 pt)
- **Sampling:** Explain the ODE solver and sampling procedure, including the number of time steps. (.05 pt)

1d. Explain the diffusion baseline

(.275 pts)

- **Mathematical formulation:** Define the implemented forward process, noise schedule, prediction target, loss, and reverse sampler; cite the relevant diffusion literature. (.125 pt)
- **Architecture and training:** Report the architecture, optimizer, and important training settings. (.05 pt)
- **Model selection:** Explain how the final model was selected, using validation-based selection or early stopping where appropriate. (.05 pt)
- **Sampling:** Explain the sampling procedure, including the number of time steps. (.05 pt)

1e. Plan the flow-matching versus diffusion comparison

(.275 pts)

- **Metrics:** Define the quality and diversity or control metrics used to compare the two models on Modality 1. Describe their implementation, sample counts, and limitations. (.175 pt)
- **Fairness:** Explain how data, conditioning, metrics, model size, training cost, sampling time or NFE, hardware, seeds, and compute are matched or disclosed. (.1 pt)

1f. Show the study overview**(.1 pts)**

Present the overview figure from the paper. It should show the establishment of the flow-matching and diffusion baselines, controllability, the contribution of the innovation, and transfer to Modality 2.

Part 2: Build the Evidence for Modality 1**2.25 points****2a. Choose the base model****(.275 pts)**

- **Section 4.2 table:** Correctly present the comparison between the trained flow-matching and diffusion baselines. **(.175 pt)**
- **Model selection:** State which model is carried forward and justify the decision from the observed trade-offs. **(.1 pt)**

2b. Present the guidance strategy and result**(.3 pts)**

- **Guidance strategy:** Define the guidance rule and target; report trained and frozen components, the strength schedule, and any normalization or clipping. **(.1 pt)**
- **Section 4.3 table:** Compare guided and unguided samples under matched settings, reporting target improvement and quality or diversity trade-offs. **(.2 pt)**

2c. Introduce the innovation**(.475 pts)**

- **Motivate the innovation:** State the hypothesis, the weakness it addresses, and how it differs from prior work. **(.2 pt)**
- **Specify the innovation:** Give the complete methodological change with equations, an algorithm, trainable and frozen components, and new hyperparameters. **(.275 pt)**

2d. Test the innovation**(.45 pts)**

- **Section 4.4 table:** Present at least two informative ablations under matched conditions, removing components or using appropriate controls for an atomic change. **(.3 pt)**
- **Conclusion from the ablations:** Explain what each comparison isolates, which components are responsible for improvement, which claims the evidence supports, and any trade-offs. **(.15 pt)**

2e. Position the method against recent work**(.75 pts)**

- **Section 4.5 table:** Benchmark against at least three relevant external methods from the last five years plus the internal baseline. Report quality, diversity or coverage, and efficiency. **(.35 pt)**
- **Fair study:** Distinguish reproduced from quoted results, report versions and uncertainty where feasible, use consistent protocols, metrics, and sample counts, and disclose material data or compute differences. **(.25 pt)**
- **Fair discussion:** Explain where the method improves, remains competitive, or performs worse, including both favorable and unfavorable outcomes. **(.15 pt)**

Part 3: Test Transfer and Close the Story**.95 points****3a. Explain the transferred method****(.3 pts)**

- **What stays the same:** Identify the part of the innovation retained in transfer to Modality 2. (.075 pt)
- **What changes:** Explain the required modality-specific adaptations. (.075 pt)
- **Training procedure:** Report the Modality 2 architecture, objective, training settings, and checkpoint selection. (.1 pt)
- **Sampling procedure:** Report the Modality 2 sampling procedure. (.05 pt)

3b. Present the transfer results (.275 pts)

- **Section 4.6 table:** Compare the transferred method against at least three relevant recent methods. Report quality, diversity or coverage, and efficiency. (.125 pt)
- **Fair study:** Distinguish reproduced from quoted results, report versions and uncertainty where feasible, use consistent protocols, metrics, and sample counts, and disclose material differences. (.075 pt)
- **Fair discussion:** Explain where the method improves, remains competitive, or performs worse, including favorable and unfavorable outcomes. (.075 pt)

3c. Decide whether the innovation transfers (.1 pts)

Compare the direction and magnitude of improvements across modalities and explain whether the evidence supports transfer.

3d. Analyze limitations and failure modes (.1 pts)

Analyze at least two meaningful limitations or failure modes, using representative samples or trajectories when informative.

3e. Conclude and identify future directions (.175 pts)

- **Connect to the study motivation:** Return to the opening hypothesis and give the strongest conclusion supported by the evidence. (.075 pt)
- **Mechanism, trade-offs, and next steps:** Interpret the mechanism and trade-offs and identify specific future experiments. (.1 pt)

Q&A

3 points

Questions will be directed to individual students and cannot be delegated to another team member. Every student should understand the whole project, not only the part they implemented or presented. This includes the datasets, preprocessing, FM and diffusion formulations, training and sampling procedures, guidance, innovation, ablations, baselines, metrics, figures, conclusions, implementation details, and Modality 2 transfer.

Words of Advice

- **Read the complete paper instructions before starting.** The later sections tell you which controls, ablations, and baselines you will need.
- **Start early.** You are training multiple generative models across two modalities, and debugging can take longer

than expected.

- **Choose feasible datasets.** A complete, well-controlled study is better than an ambitious project that never reaches interpretable results.
- **Get the base models working first.** Verify FM, diffusion, sampling, and metrics before adding the innovation.
- **Start from a hypothesis.** Decide what limitation you want to address, then design the innovation and ablation around that question.
- **Keep comparisons fair.** If a baseline cannot be run under exactly the same settings, explain the mismatch clearly.
- **Do not treat Modality 2 as an afterthought.** A weak transfer result is still useful if you explain what it reveals about the method.
- **Save configs and intermediate results.** Keep seeds, metric outputs, generated samples, and plotting data so your tables and figures can be reproduced accurately.
- **Know the papers you cite.** You may be asked about a baseline's assumptions, architecture, or evaluation during the defense.
- **Ask early if you are unsure whether an idea counts as an innovation.**

Groups

Here we provide group assignments and each member's PennKey and email, so you can easily reach each other.

Group	Student	PennKey	Email
1	Mats Hansen	matshan	matshan@sas.upenn.edu
1	Xinyue Ai	xinyueai	xinyueai@engineering.upenn.edu
1	Wenrui Ma	ma10	ma10@seas.upenn.edu
1	Binghao Yan	binghao	binghao.yan@pennmedicine.upenn.edu
2	Bobo Li	boboli	okayboboli@gmail.com
2	Henry Huang	hyhuang	hyhuang@sas.upenn.edu
2	Idea Idehpour	ididea	idepoorparsa@gmail.com
2	Haimo Fang	edfang	edfang@sas.upenn.edu
3	Audhav Durai	audhav	audhav@wharton.upenn.edu
3	Chang-Uk Jeong	changuk	chang-uk.jeong@pennmedicine.upenn.edu
3	Roshan Bellary	rbellary	rbellary@seas.upenn.edu
3	Pedram Bayat	pbayat	pbayat@seas.upenn.edu
4	Yan Pan	pan4	pan4@seas.upenn.edu
4	Yijia Jiang	jiangyij	Yijia.Jiang@Pennmedicine.upenn.edu
4	Anushka Sheoran	asheoran	asheoran@seas.upenn.edu
4	Neha Peddinti	peddinti	peddinti@seas.upenn.edu
5	Rutva Patel	rp7404	rp7404@engineering.upenn.edu

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Table 1 – continued

Group	Student	PennKey	Email
5	Ratish Panda	panda1	panda1@seas.upenn.edu
5	Darrel Dsouza	darreld	darrel.dsouza20@gmail.com
5	Pranshav Lakhia	pranshav	pranshav@engineering.upenn.edu
5	Imran Nasrullah	inasrull	inasrull@seas.upenn.edu
6	Edward Sekula	ersekula	ersekula@seas.upenn.edu
6	Edward Donson	edonson	edonson@engineering.upenn.edu
6	Jacob Yin	jacobyin	jacobyin@seas.upenn.edu
6	Kevin Chen	bchen72	bchen72@seas.upenn.edu